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United States Patent	12397242
Kind Code	B2
Date of Patent	August 26, 2025
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Track hurdle assembly

Abstract

A huddle assembly is provided having a base, two upright members, and a cross bar extending between the two upright members. The cross bar includes a first bar member and a second bar member. An attachment member temporarily secures the two bar members together to form a continuous bar. The two bar members are pivotably supported on a respective upright member. The temporary attachment of the two bar members permits their separation upon contact by a runner. The pivotable attachment to the uprights permits the rotation of the bar members, during separation, about a vertical axis of the upright members.

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Family ID:	1000008782685
Appl. No.:	18/477109
Filed:	September 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240115969 A1	Apr. 11, 2024

Related U.S. Application Data

us-provisional-application US 63414015 20221007

Publication Classification

Int. Cl.: **A63K3/04 (20060101)**

U.S. Cl.:

Field of Classification Search

CPC: A63K (3/04); A63K (3/043); A63K (3/046)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of the filing of U.S. Provisional Application No. 63/414,015, filed Oct. 7, 2022, the disclosure of which is herein incorporated by reference.

BACKGROUND

(1) An event in track and field competition is a race using hurdles that are spaced along a running track. The runner must stride over the hurdles while maintaining proper step cadence.

SUMMARY

(2) The present assembly is a track hurdle having a cross bar that separates and pivots when contacted during use. The cross bar becomes a gate that opens when contacted, allowing the runner to practice hurdling without incurring any significant impact when not completing the stride over the cross bar. The assembly is preferably portable and easily assembled.

(3) In embodiments, a hurdle assembly includes a base, two upright members and a cross bar suspended between the two uprights. Each upright member is preferably fixed on an opposite side of the base. The cross bar may include a first bar member and a second bar member each pivotably supported on a respective upright member. The two bar members are linearly aligned between their pivot mountings. A striking force on the cross bar causes a separation of the two bar members, with the each bar members pivoting away from the alignment on their upright support. The assembly may further include an attachment member that releasably secures the two bar members together. The striking force causes a separation of the attachment member from at least one bar member so as to permit the pivoting of the bar members about their respective mountings.

(4) A hurdle assembly may be provided with a fixed base, two upright members projecting

vertically from the base, and a cross bar extending horizontally between and preferably perpendicular to the upright members. The cross bar may include a first bar member and a second bar member, with the two bar members being pivotably supported on a respective upright member for rotation preferably about a vertical axis. An attachment member temporarily secures the two bar members together with a friction fit to form the linear cross bar. The temporary attachment of the two bar members permits their separation upon contact by a runner and the pivotable attachment permits the outward rotation of the bar members.

(5) The bar members of the cross bar are preferably made of a soft foam material. The upright members may include a telescoping structure, with an outer collar co-axially aligned with a post member. The post member may be fixed to the base and project vertically upward therefrom. The base may be formed from a bottom bar having a base member attached at each end thereof and a leg member projecting rearwardly, preferably perpendicular to the bottom bar. Stabilizing foot members may be provided forward of the base members, and preferably axially aligned with the leg members. Height adjustment means may be provided as part of the uprights. The height adjustment means may be frictionally engaged on a post portion of the uprights and support the telescoping collar. Movement of the adjustment mean along the post members adjust the relative height of the cross bar above the base member. The adjustment means may include a clamp structure for frictionally engaging the post member.

(6) The pivotable support may be formed by corner members fixed to a rotational collar that is telescopically formed as part of the upright members. The bar members are preferably attached to the corner members and directed horizontally, in a perpendicular relation with the uprights.

(7) The pivotable support may be formed by corner members, with one or more rotationally mounted to an upright member. The pivot mounting may be formed by an insert flange telescopically positioned within a portion of the upright members. The bar members are preferably attached to the corner members and directed horizontally, in a perpendicular relation with the uprights.

(8) The bar members are preferably attached to the corner members by a friction fit. The friction fit of bar members with the corner members is preferably being greater than the friction fit of the bar members with the attachment member.

(9) In embodiments, a gated cross bar may be fixed to the uprights of an existing hurdle assembly as a replacement for the existing fixed cross bar. The cross bar may include a first bar member and a second bar member each attached to corner pivot that may be secured to the existing uprights. The two bar members are linearly aligned between their corner pivot members. The gated assembly includes an attachment member that releasably secures the two bar members together. The bar members are preferably made of a soft foam material and frictionally engage the attachment member. The attachment member is preferably inserted into one end of each bar member and preferably positioned at the center of the cross bar. A striking force causes a separation of the attachment member from at least one bar member so as to permit the pivoting of the bar members about their respective corner pivots. The gated bar assembly permits the adaption of the separable and pivoting cross bar to existing hurdles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a perspective view of an embodiment of a hurdle assembly.

(2) FIG. 2 shows a rear elevation view of the hurdle assembly embodiment shown in FIG. 1. A number of cross section views are shown, including Section A-A as taken along line A-A, Section B-B as taken along line B-B, Section C-C as taken along line C-C, Section D-D as taken along line D-D, and Section E-E as taken along line E-E.

- (3) FIG. 3, parts (A) and (B), show a top plan view of the hurdle assembly of FIGS. 1 and 2.
- (4) FIG. 4 shows a partial rear elevation view of the hurdle assembly, with the various parts (A), (B) and (C) showing the cross bar adjusted to heights H1, H2, and H3, respectively.
- (5) FIG. 5 shows a side elevation view of the hurdle assembly, with the various parts (A), (B) and (C) showing the cross bar adjusted to the heights H1, H2, and H3, respectively
- (6) FIG. 6A shows various view of an adjustment clamp portion.
- (7) FIG. 6B shows partial views of the elements of the cross bar assembly as taken from FIGS. 3A and 3B, respectively.
- (8) FIGS. 7A, 7B and 7C show various embodiments of a cross bar pivot assembly.
- (9) FIG. 8 shows one or more dismantled hurdles stored in a travel bag.

DETAILED DESCRIPTION

(10) In the various figures, like numbers are intended to identify similar elements in the assembly. In FIG. 1 there is shown a hurdle assembly that is generally identified by the number 10. The assembly 10 is formed by a base 12, two uprights 14, 16 and a cross bar or cross bar 18. The cross bar 18 as shown includes two bar members 20, 22 that project from opposing corner members 26, 28. The corner members 26, 28 are provided respectively at the top of the uprights 14, 16. The uprights 14, 16 each include an outer collar 30 that is telescopically positioned over an inner post 32. The collar 30 preferably has an inside diameter that is greater than the outside diameter of the post 32, such that the collar 30 may both rotate about a vertical axis of the post 32 and move longitudinally up and down along the post 32. A similar collar 30 and post 32 assembly is contemplated to be provided for upright 14 on the opposite side of the assembly. In the embodiment of FIG. 1, the corner members 26, 28 are positioned at the top end of their respective uprights 14, 16, with the collar portions 30 fitting over their respective post members 32. The corner members 26, 28 are preferably pivotally secured to their respective uprights 14, 16.

(11) The base 12 of the assembly 10 is formed by a bottom bar 34 and two legs 36, 38. The legs 36, 38 project rearwardly from respective base members 44, 46 and are attached at opposite ends to the bottom bar 34. The legs 36, 38 are respectively fixed to the end member 44, 46 on opposite sides of the base 12. The legs 36, 38 and post members 32 are preferably attached to the base members 44, 46 by a friction fit. The friction fit is contemplated to provide stability to the assembly 10, while permitting relatively easy removal for dismantling/disassembly.

(12) Also illustrated in FIG. 2, the cross bar 18 is formed by two bar members 20, 22, with an attachment member 24 provided therebetween. The bar members 20, 22 are hollow, as represented by section D-D. The attachment member 24 is positioned within the hollow ends of bar members 20, 22. This relationship is shown in Section E-E and in FIG. 6B. A portion of member 24 is inserted into the center end of each bar member 20, 22 to form the cross bar 18. The attachment member 24 temporarily fixes the bar members 20, 22 together. As is discussed further below, the bar members 20, 22 may separate or split, with each bar member pivoting away about their respective uprights 14, 16. One collar portion 30 is attached to a respective corner member 26, 28. The outside ends of the bar members 18, 20 are fixed to their respective corner members 26, 28 by the flange 48. The flange 48 is inserted into the hollow of the outside ends of the respective bar member 20, 22. The inserted position of the flange 48 is shown in section C-C.

(13) In the top views of the assembly in FIG. 3, the cross bar 18 is shown in the operative position. In part (A), the two bar portions 20, 22 are fixed together by attachment member 24. The detail of the assembly is further illustrated in the cross section of FIG. 6B, part (A). The two bar portions 20, 22 are hollow, having an inner diameter. The attachment 24 is inserted into the respective (central) ends and creates a friction fit. When separated (see, e. g., part (B) of both FIGS. 3 and 6B) the attachment member 24 slides out of one of the bar portions—shown as bar 20—and remains positioned in the end of the other bar portion 22. In FIG. 3, part (A), the assembled cross bar 18 is fixed to the uprights (14, 16) by corner members 26, 28. The corner members 26, 28 include the projecting flange 48 that fits within the respective (outer) ends of the bar members 20, 22. See also

the partial view of FIG. 7A. The corner members **26**, **28** are rotational mounted on the top end of their respective uprights **14**, **16**, e. g., as shown in FIGS. 1 and 2.

(14) In FIG. 3, part (B), there is shown an arrow simulating a force being applied to the cross bar **18** causing a separation of the bar portions **20**, **22** and a rotation of the bar portions about the axis of the associated upright **14**, **16**. The force arrow represents a runner striking the cross bar **18** in an attempt to clear the hurdle assembly **10**. Due to the friction fit of the attachment member **24** within the hollow ends of the bar members **20**, **22**, there is a relatively easy separation and pivoting of the two bars. Hence, the runner is able to practice with minimal resistance being applied against the force (arrow) of the strike. Referring again to FIG. 7A, the pivot of the bar member **22** occurs by its attachment to the corner member **26**, which is fixed to the outer collar **30** and which is in turn slidably positioned over the fixed post member **32**. The collar **30** and post **32** are axially aligned and the inner diameter of the collar being greater than the outer diameter of the post **32**. A similar arrangement for upright **14** with its corner member **28** is also contemplated.

(15) As shown in FIG. 3, parts (A) and (B), forward projecting foot members **54**, **56** are fixed to the respective base members **44**, **46**. The foot members **54**, **56** are positioned opposite of the leg member **36**, **38** on the base **12** of the assembly. The foot members **54**, **56** are considered optional and serve to add resistance to forward movement and/or tilting of the assembly **10** during a striking force (e.g., the arrow in FIG. 3, part (B)). The foot members **54**, **56** may also provide additional stability in windy conditions.

(16) In FIGS. 4 and 5 there is shown height adjustment of the uprights **14**, **16**, to raise or lower the cross bar **18**. The length of the uprights **14**, **16**, and thus the height of the cross bar **18**, may be changed to match, for example, the age, height or skill level of the runner. Changing the height of the cross bar **18** is created by the movement of an adjustment member **42** on the post members **32**. As shown in the figures, the adjustment member **42** is attached to the post member **32** at a position above the base member **46**. The adjustment member **42** is shown in more detail in FIG. 6A as having an open c-shaped clamp portion **51**. Clamp portion **51** is sized to engage with the post member **32**. An opposing nub **52** is shown and may be provided so as to assist in the sliding movement of the adjustment member **42** along the post **32**.

(17) Referring to FIGS. 4 and 5, parts (A) thereof, the clamp **51** engages the post **32** and is set at a specific position, preferably marked by a reference line **50**. A similar setting is provided for the adjustment member **42** on post **32** of upright **14**. The bottom end of the collar **30** rests on adjustment member **42**, which is fixed to the post **32**. This arrangement in turn fixes the cross bar **18** to a height H1. In part (B) of these figures the member **42** is raised upwardly on the post **32** to a second reference line **50**. This adjustment raises the cross bar **18** to a new height H2. In part (C), the adjustment member **42** is again raised to a third reference line **50** on the post **32** to push the collar **30** up further and set the cross bar to height H3.

(18) In use, it is contemplated that the frictional engagement of the adjustment member **42** on the fixed post **32** is greater than the frictional engagement of the attachment member **24** within the hollow of the two bar portions **20**, **22**. The adjustment member **42** must support the weight of the collar, corner member and cross bar and maintain that position at the desired height. However, the engagement preferably permits a sliding adjustment along the post. Other adjustment control members and assemblies are possible.

(19) Referring again to FIG. 7A, the pivot of the bar portion **22** occurs by its attachment to the corner member **26**. The pivotal attachment of the corner member is formed by it being fixed to the outer collar **30**. The collar is telescopically positioned over the post member **32** and thus may rotate about the vertical axis of the upright, which is formed by the post member **32**. The dimensional relationship between collar **30** and post **32** allow the collar to rotate about the post **32**.

(20) In FIGS. 7B and 7C, there is shown a potential alternate structure for the upright **14'** and corner member **26'**. As shown, the corner member **26'** includes a lateral flange **48A**, which is similar to the flange **48** in FIG. 7A. The lateral flange **48A** is inserted into the interior of the bar

member **22**. The pivot attachment for the corner member **26'** may be formed an insert flange **48C** that is fixed to the corner member **26'**. A projection **48B** extends from the insert **48C** in a downward direction. As shown, the outer dimension of the projection **48B** is contemplated to be less than the inside dimension of the post **32**. In FIG. 7B the projection **48B** is positioned inside the post **30** and may rotate about the vertical axis of the post **32**. The outer collar **30** abuts against the bottom surface of the corner member **26'** and/or a shoulder formed by the insert **48C**. In this arrangement, the corner member **26'** may rotate about the axis of the upright. Along with the attached bar portion **22**. The rotation of the corner member is independent of the collar **30** and the post **32**. In FIG. 7B, the relative positioning of the parts is contemplated to correspond to the height H1 of FIGS. 4 and 5, parts (A). In FIG. 7C, the height of the cross bar is raised, such as to H2 of FIGS. 4 and 5, parts (B). The collar **30** has now moved vertically up the post **32** and there is a separation of the projection **48B** from the top end of the post **32**. In this FIG. 7C, the projection **48B** continues to serve as the pivot attachment for the corner member and its associated bar portion.

(21) In FIG. 7C, the corner member **26'** is free to rotate either with the collar **30** or separately therefrom. It is contemplated that the arrangement of FIGS. 7A and 7B may be adapted for use in other upright and/or based structures. For example, in an existing hurdle structure, the fixed cross bar may be removed and the separable cross bar **18** may be attached to the existing uprights, with the corner member **26'** positioning the cross bar **18** and permitting rotation. The dimensions of the flange member **48B** and/or insert **48C** may need to be conformed to the upright structure of the existing hurdle. An adaptor structure may also be added to accommodate different dimensions for the existing structures and to permit the corner member/cross bar combination to be moved from the assembly **10** to an existing structure.

(22) As shown in FIG. 8, the hurdle **10** may be dismantled/disassembled for storage and transport. As shown, preferably, one or more assemblies will fit within a small to medium duffle bag **40**. The assembly is contemplated to be formed by frictional engagement of the parts. For example, the bottom bar **34** is inserted at opposite ends of the base members **44**, **46**. Leg members **36**, **38** are also inserted into the respective base members **44**, **46**. If provided, foot members **54**, **56** are inserted into the from end of the base members **44**, **46**. This sub-assembly forms the base **12** of the hurdle. To form the uprights **14**, **16**, post members **32** are inserted into and project upwardly from the base members **44**, **46**. Each collar **30** is slid over its respective post member **32** and may slide linearly along and rotate about the post member. A key or frictional element may be added to the overlap of the collars on the posts to restrict separation during, for example, movement of the hurdles from one position or location to another. In the embodiment of FIG. 7A, the collars are attached to their respective corner members **26**, **28**. The attachment may be a friction fit or may be fixed in other ways. The attachment connects the movement of the corner with that of its associated collar. As discussed previously, the assembly of the corners and uprights in the embodiment of FIGS. 7B and 7C is different, but still permits the bar members to rotate about the axis of the uprights, as defined by the post members **32**. The length of the uprights may be defined by an adjustment member fit onto the post member. The adjustment member is secured to the post and supports the collar member, which sit on top of the adjustment member. The cross bar **18** is assembled from the two bar member portions **20**, **22**. One end of the bar member **20** is slid over a projecting flange **48** of the corner member **28**. The second bar member **22** is similarly assembled. The two bar members **20**, **22** are linearly aligned between the two corner members **26**, **28**. An attachment member **24** may be inserted into the center ends of the two bar members **20**, **22** to form a releasable connection between the two bar portions.

(23) In embodiments, the hurdle assembly may be constructed of common, relatively rigid, plastic pipe members, such a piping made from polyvinyl chloride (PVC) or other common materials. The cross bar may be formed by a foam pipe insulation material, made, for example, of a polyethylene material. Other materials and structures are possible, including polyethylene foam "pool noodles". preferably, the cross bar is made of a relative soft material, having sufficient stiffness and strength

to form a linearly strait bar that extends between the uprights and that may withstand repeated contact during use.

(24) The relative dimensions and frictional contact between the various members of the assembly may be adjusted for desired operation. For example, the bar members are attached to their respective corner members by the projecting flanges. The flanges are contemplated to be relatively longer than the central attachment member and may be inserted into the ends of the bar portions by a greater distance. For example, the flanges may extend approximately 2.5 inches from the corner holder, while the attachment member preferably extends approximately 0.5 inches into the ends of the two bar portions. Adjustments to the length and material of these elements may be made depending on the dimensions, physical attributes, and frictional properties of the relevant parts. This relationship leads to the bar portions to be temporarily fixed to one another and to easily separated at the center when being subject to a transverse or downward force during use.

(25) Typically, a series of hurdle assemblies are lined up on a track in a spaced relationship. The runner is required to sprint along the track lane and jump over the hurdles in succession. The assembly as described herein provides the advantage of the cross bar **18** separating should the runner fail to clear a hurdle. The separation is preferably created with minimal transverse or downward contact and without abrasion being caused to the runner. The pivot around the fixed post members of each upright further adds to the ability of the bar portions to separate and pivot away from the center. Alternated adjustment mechanisms may also be included. A resilient return mechanism may be added for the automatic reset of the position of the bar members. Alternate forms of the attachment member may be used.

(26) Other features and advantage of the assembly may become apparent by a review of the description above and the included illustrations.

Claims

1. A hurdle assembly comprising: a base, two upright members, each respective upright member fixed on opposite sides of the base, a cross bar suspended between the two upright members, the cross bar having a first bar member and a second bar member, corner members, each pivotably supporting one of the bar members on a respective upright member, the first and second bar members positioned substantially perpendicular to the respective upright member, and a bar attachment releasably securing the first and second bar members together and in an end-to-end alignment, wherein a striking force on the cross bar causes a release of the bar attachment and a separation of the two bar members, with the two bar members pivoting on their corner member about a vertical axis of their respective upright members; wherein the first and second bar members are made of a soft foam material, and wherein the first and second bar members have a hollow core and the bar attachment is positioned within the hollow core of each of the first and second bar members.
2. The hurdle assembly of claim 1, wherein each respective upright is fixed on an opposite side of the base.
3. The hurdle assembly as in claim 1, wherein the upright members project substantially vertically from the base.
4. The hurdle assembly as in claim 1, wherein the corner members comprise a telescoping structure having an outer collar, wherein the upright members each include an inner post member, and wherein the outer collar is co-axially aligned with an inner post member.
5. The hurdle assembly of claim 4, wherein the inner post member of the upright members are fixed to the base, and wherein the base comprises a transverse bottom bar, a joint member attached at each end of the bottom bar and a leg member projecting rearwardly from each respective joint member.
6. The hurdle assembly of claim 5, wherein the base further comprises a foot member attached to at

least one joint member, the foot member projecting forward of the corresponding base member.

7. The hurdle member of claim 4, further comprising a height adjustment member provided on at least one of the upright members, the height adjustment member frictionally engaged on an outer surface of the inner post member of the upright member, the height adjustment member in contact with and vertically supporting the telescoping outer collar of the corner member, the frictional engagement of the height adjustment member being releasable for adjusting the vertical position along the upright member.

8. The hurdle member of claim 4, wherein each corner member is fixed to the outer collar, with the outer collar capable of rotation about the vertical axis of the upright members formed by the corresponding post member.

9. The hurdle assembly as in claim 4, wherein the telescoping structure further comprises: a friction element formed between each of the outer telescoping collars and inner post members, the friction element restricting separation of each respective collar and inner post member.

10. The hurdle assembly of claim 1, wherein each corner member is mounted on the respective upright member by a flange telescopically positioned with the upright member, and wherein the flange is capable of rotation about the vertical axis of the upright member.

11. The hurdle assembly of claim 1, wherein the respective first and second bar members are attached to the corresponding corner member by a friction fit.

12. The hurdle assembly as in claim 11, wherein the bar attachment comprises an insert positioned within projected ends and frictionally retained by the hollow core of the first and second bar members, and wherein the friction fit of the bar members with the corner members is greater than the frictional securing of the insert within the hollow core of the bar members.

13. A hurdle assembly comprising: a base, two upright members, each upright member vertically positioned on opposite sides of the base, a pair of corner pivot members, the corner pivot members positioning a respective bar member on a respective upright member, each corner pivot member comprising a telescoping structure having an outer collar co-axially aligned with an inner post member formed as part of the upright member, the coaxially aligned outer collar rotationally fixing the corner pivot member to the respective upright member, and wherein the outer collar is rotational about the vertical axis of the upright member, a cross bar suspended between the two upright members, the cross bar having a first bar member and a second bar member, one bar member fixed to one of the corner pivot members and pivotably supported on a respective upright member by the collar member of the corner pivot member, and the first and second bar members linearly aligned between their respective pivotal supports; a bar attachment positioned on projected ends of the two bar members, the bar attachment releasably fixing the projected ends together and creating the linearly alignment of the first and second bar members, wherein a striking force on the cross bar causes a release of the bar attachment and a separation of the two bar members, and each bar member freely rotating in a horizontal plane about the vertical axis of their respective upright member.

14. The hurdle assembly as in claim 13, wherein the bar attachment comprises an insert positioned on the projected ends of the first and second bar members, the insert frictionally engaged with at least one of the two bar members to releasably fix the projected ends of the bar members.

15. The hurdle assembly as in claim 14, wherein the first and second bar members comprise a hollow core and the insert frictionally positioned within the hollow core of the at least one bar member.

16. The hurdle assembly as in claim 15, wherein the first and second bar members are made of a soft foam material.

17. The hurdle assembly of claim 13, further comprising a friction element formed between each of the telescoping collars and posts, the friction element restricting separation of the respective collars and posts.

18. A cross bar assembly to be attached to uprights of an existing hurdle assembly, the cross bar

assembly comprising: a first bar member and a second bar member, a pair of corner members, one bar member attached to one respective corner member, wherein the bar member projects horizontally from the corner member, a bar attachment positioned on projected ends of the two bar members and releasably securing the two bar members in an end-to-end linear alignment, between the two corner members, and a corner pivot formed as part of each of the corner members, the corner pivot rotationally mounting the at least one corner member to the upright member of the existing hurdle assembly, such that the corner member is capable of rotation about a vertical axis formed by the upright member of the existing hurdle assembly, the corner pivot including a collar telescopically fixing the corner member to the respective upright member of the existing hurdle assembly for the rotation about the vertical axis of the upright member, wherein a striking force on the releasably secured bar members causes a release of the of the bar attachment and a free rotation of the corner pivot about the vertical axis of the upright member of the existing hurdle assembly and a rotation of the bar members in a horizontal plane.

19. The cross bar assembly as in claim 18, wherein the bar members have a hollow tubular form.

20. The cross bar assembly as in claim 19, wherein the bar members are made of a soft foam material.

21. The cross bar assembly as in claim 19, wherein the bar attachment is inserted into and frictionally secured within the hollow of the projected ends of each bar member.

22. The cross bar assembly as in claim 18, wherein the bar attachment is frictionally engaged within the two bar members to releasably fix the projected ends of the first and second bar members.
